

CSMMetal20250004 — EMF Protection Color Scheme Derivation V2.0

Chromatic Defense: Harmonic Engineering of Electromagnetic Shielding Materials

Version 2.0 — Revised & Expanded | June 2026

(Duplicate of CSMMetal folder version; canonical reference copy in CSMAegisHome000000000001)

Δ Change Log

- Curie temperature data corrected with 2025 precision magnetometry
- Octave-scaling method clarified as heuristic mapping (not fundamental physics)
- Blackbody Wien’s Law derivation added for rigorous color-from-temperature correlation
- Extended palette: 8 colors with CIE Lab* manufacturing specifications
- MXene FSS V2.0 layer integrated into wall assembly diagram

Core Palette Summary

Color	Hex	Physical Basis	Application
Stellar Aegis Blue	#1B4F72	YInMn Blue, Mn ³⁺ crystal field	Primary exterior, NIR rejection
Cobalt Forge Orange	#FF6500	Cobalt blackbody at 1388K Curie T	Heat warning (thermochromic)
Iron Carrington Red	#C0392B	Iron blackbody near 1043K Curie T	GIC alert indicator
Nickel Threshold Amber	#D68910	Nickel Curie T = 627K	Moderate heat warning
Schumann Earth Green	#27AE60	56th octave of 7.83 Hz	Interior walls, calming
Argentum Cyan	#1ABC9C	CoAl ₂ O ₄ spinel reflectance	Sterile surfaces, laser counter
Void Black	#1C2833	Ni-coated microsphere absorber	RF absorption patches
Phoenix White	#FDFEFE	SiO ₂ aerogel/TiO ₂	Thermal insulation indicator

See CSMMetal20250004 V2.0 in the CSMMetal folder for full derivation.